

Roton-like softening of a transverse photon branch in quantum spin ice

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Quantum spin ice has a well-established long-wavelength description in terms of an emergent photon, and previous exact-diagonalization and semiclassical work already establishes that the underlying compact gauge theory is strongly interacting. What those results do not fix is how the microscopic ring model distributes its longitudinal spectral weight at finite momentum. We compute the zero-temperature correlation function $S^{zz}(\mathbf{q}, \omega)$ of the local Ising pseudospin, which is exactly the emergent electric field, in the ground-state winding sector of 48-, 64- and 72-site pyrochlore clusters, resolving every eigenstate and matrix element rather than a broadened lineshape. The response is photon-like at the smallest accessible momentum and reconstructs at the zone boundary. At L a collective level appears at $1.02K$ on 48 sites, $0.95K$ on 64 and $1.07K$ on 72, carries at least half the weight at that momentum, and lies near half the Gaussian-photon energy. A basis-independent polarization analysis identifies it as the lower of the two transverse branches rather than a new species, and few-photon kinematics place it a factor of five below the three-photon threshold. We call the mode roton-like in the Feynman sense: its softening tracks the ratio of oscillator strength to equal-time structure factor, $f_1(\mathbf{q})/S(\mathbf{q})$, both of which are ground-state quantities, and the ratio correctly predicts in advance which of two inequivalent L -type momentum classes on the 64-site cluster reconstructs. Two microscopic diagnostics separate the level from an ordinary dressed photon: a Rokhsar–Kivelson flippability potential, used only as an infinitesimal probe at $V = 0$, shows that it gains flippable hexagons while losing ring-exchange coherence; and an exact plaquette decomposition attributes 71% of its variational relaxation to the one hexagon family that the electric-field wave leaves entirely unconstrained, a lattice analogue of backflow. The same reconstruction appears as an order-one polarization splitting at the zone boundary, and a further 16–38% of the weight lies in levels absent from the Gaussian one-photon theory. We state explicitly which conclusions are limited by the accessible cluster sizes.

I. INTRODUCTION

Quantum spin ice realizes a compact $U(1)$ gauge theory inside a frustrated magnet. In the two-in-two-out manifold of the pyrochlore lattice, virtual exchange flips the six spins around an elementary hexagon [1, 2]. The resulting deconfined phase supports a gapless emergent photon together with gapped electric and magnetic charges [3–6]. Because the local moment is the emergent electric field, $S^z = E$, magnetic neutron scattering measures electric-field fluctuations through $S^{zz}(\mathbf{q}, \omega)$. The long-wavelength photon and its characteristic correlations have been established by gauge theory, quantum Monte Carlo, and constrained-space numerics [2, 7–10] and motivate searches in a growing set of candidate materials [11–19].

The microscopic problem is nevertheless not weakly interacting. Gaussian treatments expand the compact magnetic energy about small flux and give a non-compact Maxwell theory whose neutron response is concentrated in one-photon poles [2, 7, 20, 21]. This description is controlled in the infrared, but several results already show that it becomes incomplete at shorter wavelength. The emergent fine-structure constant is large [10]; semiclassical expansions find strong anharmonic renormalization

and, in particular, an interaction-induced splitting of the two transverse photon polarizations at intermediate wave vector [22–24]; and quantum Monte Carlo with analytic continuation finds substantial zone-boundary broadening not captured by Gaussian electrodynamics [25]. Constrained-space exact diagonalization has been pushed to larger clusters than ours — to 96 spins in Ref. [10] — but for a different purpose: extracting ground-state and low-energy parameters of the emergent electrodynamics, for which one does not need the full set of eigenstates and matrix elements. Resolving the operator content of the spectrum is more expensive per spin than locating its lowest levels, and that is the trade we make. Within that more demanding calculation we reach 72 spins, which required abandoning the single-word bit-mask representation that caps such work at 64.

The unresolved question we address is therefore narrower: *how is the longitudinal spectral weight of the microscopic ring model distributed among its finite-energy eigenstates at finite momentum?* Analytic continuation can reveal broad dynamical features but not the individual finite-size poles, matrix elements, and state character needed to answer this question sharply, while previous exact diagonalization was organized around low-energy QED parameters rather than the complete operator-resolved response. We compute the matrix elements of $S^z(\mathbf{q})$ directly in the constrained Hilbert space. The result is a long-wavelength photon that survives as expected and a zone-boundary spectrum that is strongly reconstructed. The most prominent feature is an anoma-

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lously soft L -point level carrying about half of the S^z spectral weight. We will call it *roton-like*: this terminology does not claim that our sparse finite-size momenta establish a local minimum of a thermodynamic dispersion. It denotes a finite-momentum collective continuation of a photon branch whose softening follows the Feynman ratio f_1/S and whose microscopic response differs qualitatively from that of the long-wavelength photon.

II. MODEL, OBSERVABLE, AND SCOPE

We study the standard pyrochlore ring-exchange Hamiltonian within the spin-ice manifold [2, 8],

$$H = -K \sum_p (W_p + W_p^\dagger), \quad W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-, \quad (1)$$

summed over contractible elementary hexagons. W_p reverses an alternating six-spin configuration and annihilates a nonflippable one. Thus the kinetic energy rewards coherent resonance among flippable hexagons. In gauge language $W_p = e^{iB_p}$ and Eq. (1) is the compact magnetic energy $-2K \sum_p \cos B_p$.

For spin one-half there is no independent bare electric stiffness within the ice manifold. In the conventional compact-gauge Hamiltonian $\frac{U}{2} \sum_\ell E_\ell^2 - K \sum_p \cos B_p$, the identification $E_\ell = S_\ell^z = \pm \frac{1}{2}$ makes $E_\ell^2 = \frac{1}{4}$ a constant. The photon stiffness is therefore generated dynamically by the ring term itself [2, 7, 22]. This is the strongly interacting limit in which we ask how the microscopic spectral weight departs from Gaussian electrodynamics.

A. The computed observable

Throughout, the quantity we compute is the zero-temperature, sublattice-summed longitudinal correlation function of the local Ising pseudospin,

$$S^{zz}(\mathbf{q}, \omega) = \frac{1}{N} \sum_\mu \sum_n |\langle n | S_\mu^z(\mathbf{q}) | 0 \rangle|^2 \delta(\omega - \omega_n), \quad (2)$$

with $S_\mu^z(\mathbf{q}) = \sum_{i \in \mu} e^{-i\mathbf{q} \cdot \mathbf{r}_i} S_i^z$ and μ running over the four pyrochlore sublattices. Here S_i^z is the component along the local $\langle 111 \rangle$ easy axis at site i , which is the component that equals the emergent electric field, $E_i = S_i^z$. All intensities quoted below — “share of the intensity,” “off-pole weight” — are fractions of $\sum_n \sum_\mu |\langle n | S_\mu^z(\mathbf{q}) | 0 \rangle|^2$ at the same $\mathbf{q}$, so they are normalized within Eq. (2) itself.

Equation (2) is not the neutron cross-section. The magnetic neutron cross-section of a non-Kramers or dipolar-octupolar spin ice involves, in addition, the ionic form factor, the g -tensor projecting the local pseudospin onto the laboratory magnetic dipole, the rotation from the four local $\langle 111 \rangle$ frames to the global frame, and the neutron polarization factor $\delta_{\alpha\beta} - \hat{q}_\alpha \hat{q}_\beta$. Those factors multiply Eq. (2) with sublattice-dependent and $\mathbf{q}$ -dependent coefficients and are material specific; for the

dipolar case they are given explicitly in Refs. [7, 26]. We do not apply them, because every statement we make is a comparison between eigenstates *at the same $\mathbf{q}$ and in the same sublattice channel*, where these prefactors are common and cancel from the ratios. They do not cancel from a comparison between different $\mathbf{q}$, and we therefore make no claim about the absolute momentum dependence of neutron intensity. When we write that a level “carries half the intensity” we mean half of the longitudinal pseudospin weight of Eq. (2) at that momentum. Converting to a measured cross-section for a specific material requires the material’s g -tensor and is outside the scope of the pure ring model.

Our calculation is exact within the specified finite Hilbert spaces, not within the full microscopic spin model. Equation (1) conserves winding, and we work in the ground-state winding sector of the spin-ice manifold. Electric charges (spinons) are absent by construction. The magnetic sector is retained, so spectral weight outside the photon branches could in principle have magnetic rather than multiphoton character; we distinguish these possibilities below. A word on which plaquettes enter Eq. (1). A periodic cluster contains six-site closed loops of two kinds: the elementary hexagons of the infinite pyrochlore lattice, and loops that close only by winding around the torus. The latter are not plaquettes of the lattice, and including them adds ring terms with no counterpart in the thermodynamic limit; on small clusters they are the majority of all six-cycles found by a naive search. We therefore keep only loops of zero winding number, of which the 48-site cluster has exactly 48, twelve in each of the four $\langle 111 \rangle$ orientations. This is a property of the cluster geometry and is checked directly on each cluster used here; no result below depends on it being taken on faith.

III. OPERATOR-RESOLVED SPECTRUM

The probe operator has an especially direct gauge interpretation. Since $S^z = E$, the state $S^z(\mathbf{q})|0\rangle$ is the electric-field wave launched by the probe. In Gaussian electrodynamics this state lies entirely in the one-photon sector. In the compact microscopic model it can overlap with every odd-parity eigenstate allowed by the same quantum numbers. Resolving those overlaps, rather than only the lowest energy in each momentum sector, is the central numerical task here.

The two gauge fields also have transparent spin meanings. E_ℓ is the Ising moment on bond ℓ , whereas B_p is the flux through a hexagon. The expectation value $\langle W_p + W_p^\dagger \rangle = 2\langle \cos B_p \rangle$ measures the coherence of the six-spin ring resonance. Since Eq. (1) contains the entire nonconstant energy in the ice manifold, changes in this coherence provide a direct energetic diagnostic of the excited states.

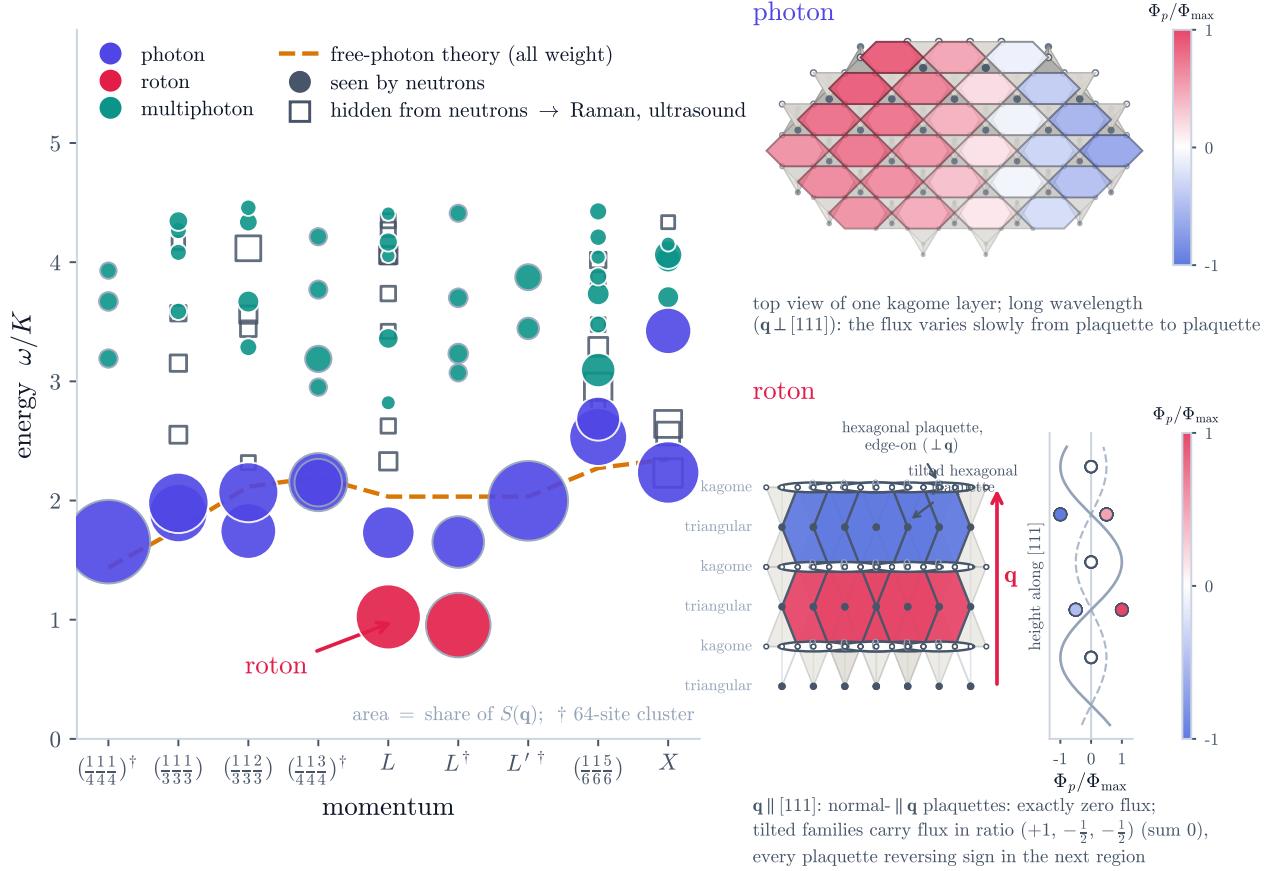


FIG. 1. Operator-resolved spectrum of the ring model in the ground-state winding sector and diagnostics of the dominant modes. Left: S^z -active levels at nine inequivalent momenta (48-site cluster; ‡ marks momenta from the 64-site cluster). Marker area is the fraction of the $S^z(\mathbf{q})$ spectral weight at that momentum carried by the level. Open squares denote spin-reversal-even levels, which are exactly dark to S^z and are shown for comparison with bilinear probes. The two photon descendants are identified by projection onto the transverse Gaussian polarizations. The anomalous L -point level is distinguished by two separately computed properties: its energy is about half the Gaussian-photon value and its flippability response is positive, $\Delta\langle n^{\text{flip}} \rangle = +0.34$, whereas the ordinary photon descendants do not appreciably gain flippable hexagons. At L it appears at $1.02K$ with 51% of the S^z spectral weight on 48 sites and at $L^\dagger$ at $0.95K$ with 50% on 64 sites. The inequivalent L' class of the anisotropic 64-site box remains photon-dominated when its equal-time structure factor is suppressed. The dashed curve is the Gaussian photon evaluated on the same lattice. Right: flux patterns associated with a long-wavelength photon and the shortest sampled wave along $[111]$. At the latter momentum, the hexagons lying in wavefronts enclose exactly zero flux; the three tilted families carry flux in the ratio $(+1, -\frac{1}{2}, -\frac{1}{2})$ and reverse sign between adjacent interlayer regions. Higher S^z -active levels carry spectral weight absent from the Gaussian one-photon description.

IV. THE ROKHSAR–KIVELSON DEFORMATION AS AN EXCITED-STATE PROBE

To classify the eigenstates we use the conventional Rokhsar–Kivelson extension of the pyrochlore ring model,

$$H(V) = -K \sum_p (W_p + W_p^\dagger) + V \sum_p n_p^{\text{flip}}, \quad (3)$$

where n_p^{flip} projects onto an alternating hexagon, so that $\sum_p n_p^{\text{flip}}$ simply counts the flippable plaquettes in a configuration. The sign convention matters for reading the

figures and we state it once: $V > 0$ penalizes flippable plaquettes, $V < 0$ rewards them, and the exactly solvable Rokhsar–Kivelson point sits at $V = +K$, where the equal-amplitude superposition is an exact ground state. Our scans run over $V/K \in [-0.5, +0.5]$ unless stated otherwise, and the anomalous level softens toward negative V , that is, toward configurations with more flippable plaquettes. This flippability potential was introduced for the pyrochlore ring model in the original gauge-theory construction and subsequently used to map the quantum-ice phase diagram and tune the emergent electrodynamics [2, 8, 10, 27]. Our use of it is different. We remain at the physical ring point $V = 0$ and use only the infinitesimal derivative of an individual excitation energy. Hellmann–

Feynman gives

$$\frac{d\omega_n}{dV} = \langle n | \sum_p n_p^{\text{flip}} | n \rangle - \langle 0 | \sum_p n_p^{\text{flip}} | 0 \rangle, \quad (4)$$

so the slope is an exact state-resolved census of the number of flippable hexagons added or removed by the excitation. No parameter is fit to obtain this diagnostic.

At L the two strongest S^z -active levels respond with opposite signs. The level at $1.73K$, close to the Gaussian-photon energy $2.03K$, loses flippable hexagons as V is increased and behaves as an ordinary dressed photon. The lower level at $1.02K$ carries 51% of the S^z spectral weight and gains flippable hexagons. It is also the lowest eigenvalue in the fully diagonalized 9104-dimensional 48-site ground-state winding sector, including both spin-reversal parities. We do not elevate that finite-sector statement to a claim about all winding sectors or about the full microscopic spin model. Parity excludes only even photon number; the three-photon channel is excluded separately, and by a wide margin, on kinematic grounds (Sec. VIII), while the flux-correlation diagnostic discussed below disfavors a localized magnetic pair. The combination of collective ancestry, strong finite-momentum softening, and qualitatively different microscopic response motivates the roton-like terminology.

V. OSCILLATOR STRENGTH AND THE FEYNMAN RATIO

The first moment of the neutron structure factor can be evaluated exactly in the ring model:

$$f_1(\mathbf{q}) = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu |g_p^\mu(\mathbf{q})|^2, \quad (5)$$

where $g_p^\mu(\mathbf{q})$ is the lattice curl, equivalently the flux driven through hexagon p by the electric-field wave. Equation (5) has the form of a Maxwell magnetic energy: the geometric factor is the squared curl of the wave, while the local stiffness is the measured ground-state ring coherence $\langle W_p + W_p^\dagger \rangle$. It contains no phenomenological stiffness parameter, reproduces the exact spectral first moment at every sampled momentum, and satisfies $\sum_p \langle W_p + W_p^\dagger \rangle = -E_0/K$ identically.

The normalized single-mode state proportional to $S^z(\mathbf{q})|0\rangle$ therefore has variational energy

$$\omega_{\text{SMA}}(\mathbf{q}) = \frac{f_1(\mathbf{q})}{S(\mathbf{q})}, \quad (6)$$

with $S(\mathbf{q})$ the equal-time structure factor. This is the standard Feynman logic [28]: a collective mode is inexpensive where its oscillator strength is small and the corresponding static fluctuation is already large. The logic itself is not new; it is also familiar from Rokhsar–Kivelson quantum dimer models [27, 29]. What is specific here is the microscopic origin of both factors at the physical $V = 0$ ring point.

A. What the ratio does and does not establish

It is essential to be clear about the logical status of Eq. (6). The ratio f_1/S is, identically, the first moment of $S^{zz}(\mathbf{q}, \omega)$ divided by its zeroth moment: it is the *centroid* of the spectrum created by $S^z(\mathbf{q})$. Observing after the fact that the centroid is low where a heavy low-lying pole exists is therefore not evidence for anything; it is a restatement. We do not use it that way.

The content of Eq. (6) is that both of its factors are properties of the *ground state alone*. f_1 follows from the double commutator and is fixed by the ground-state plaquette coherences $\langle W_p + W_p^\dagger \rangle$ and lattice geometry; $S(\mathbf{q})$ is the equal-time correlator. Neither requires knowledge of any excited state. The ratio is thus usable as a predictor, evaluated before the spectrum is computed, and the meaningful tests are those in which it predicts something not already known.

Two such tests are available here, and both are reported below. The first is a class prediction: the anisotropic 64-site cluster contains two inequivalent L -type momentum classes with substantially different $S(\mathbf{q})$, and the ground-state ratio selects, in advance, which of the two should reconstruct. The second is the trajectory under the Rokhsar–Kivelson bias: $f_1(L)$ is constant to better than one percent while $S(L)$ grows and the level softens, so the ground-state factor moves first and the spectral response follows. Statements of the form “the centroid is low at L ” carry no weight on their own and are not used as evidence in what follows.

VI. WHY THE SOFTENING OCCURS AT L

Two ingredients coincide at this momentum. The first is geometric. A hexagon contained in a wavefront, i.e. in a plane of constant phase perpendicular to $\mathbf{q}$, encloses zero flux from the electric-field wave: the alternating loop sum defining g_p vanishes exactly. Such a plaquette contributes nothing to Eq. (5). Pyrochlore hexagons are perpendicular to the four $\langle 111 \rangle$ directions, and at the sampled zone-boundary point $L \parallel [111]$ one of the four hexagon families lies in the wavefronts. That family therefore drops out of the oscillator-strength cost exactly, and the vanishing is an identity, not a small number: resolved by family, the geometric factor $\sum_p \sum_\mu |g_p^\mu(\mathbf{q})|^2$ at L is (96, 96, 96, 0), against (96, 96, 96, 96) at X . The reduction of the *geometric* factor is therefore exactly 3/4. The reduction of f_1 itself is not exactly 3/4, because Eq. (5) weights each plaquette by its own ring coherence $w_p = \langle W_p + W_p^\dagger \rangle$, which is not uniform on an anisotropic cluster; numerically $f_1(L)/f_1(X) = 0.684/0.880 = 0.777$.

It is important that this geometric decoupling is necessary but not sufficient. It occurs at every momentum along $[111]$: at the smallest such momentum the family resolution is (72, 0, 72, 72), again with one family exactly dark. Yet that momentum shows no anomaly, its dominant level sitting at $1.98K$ against a Gaussian value of

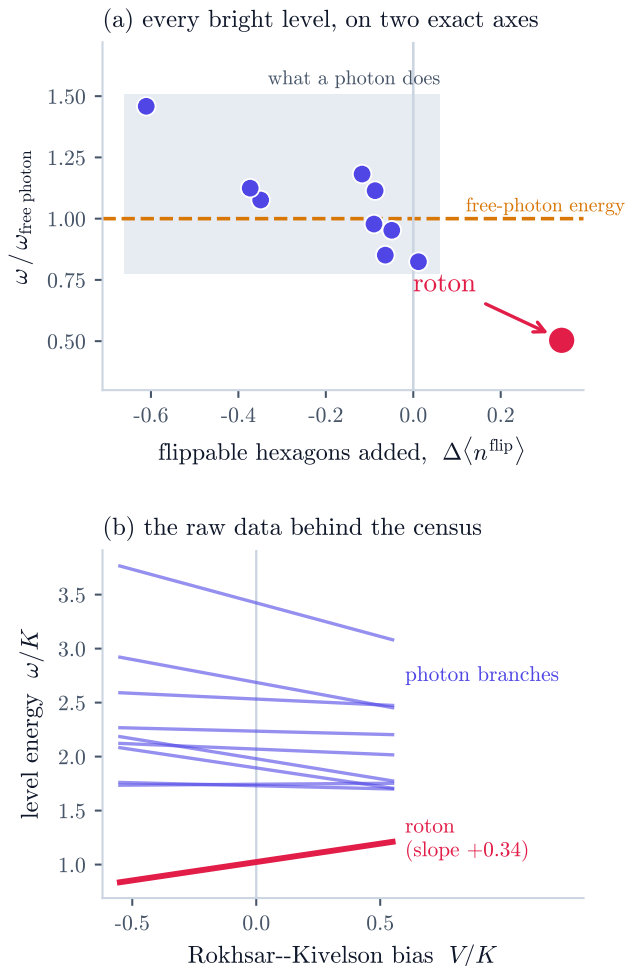


FIG. 2. State-resolved diagnostic of the anomalous L -point mode. (a) Every S^z -active 48-site level is placed on two separately computed axes. The vertical coordinate is its energy divided by the Gaussian-photon energy at the same momentum, which involves one fitted overall scale and no other parameter (Appendix A). The horizontal coordinate is the Hellmann–Feynman flippability census of Eq. (4). The photon descendants lie within 0.82–1.46 of the Gaussian energy and gain at most 0.01 flippable hexagons. The L -point roton-like level is anomalous on both axes: approximately one-half of the Gaussian energy and $\Delta \langle n^{\text{flip}} \rangle = +0.34$. (b) Evolution under the standard RK bias V . The plotted census is evaluated exactly at $V = 0$; finite-difference slopes agree to about 0.01 away from level crossings.

1.76K. What distinguishes L is the second ingredient below.

The second ingredient is dynamical. In the interacting $V = 0$ ground state the equal-time structure factor is largest at L among the sampled momenta: $S(L) = 0.370$, compared with 0.19–0.29 elsewhere. The combination of reduced f_1 and enhanced S makes f_1/S smallest at L among the twelve cluster momenta and makes it the only sampled point where the single-mode energy falls below the Gaussian-photon line.

Crucially, the enhancement of $S(L)$ is not inherited from the ice constraint alone, and two independent controls establish this.

The first is exact and requires no sampling. The equal-amplitude superposition over the *same* 9104 configurations of the *same* cluster is the classical ice ensemble restricted to exactly what the quantum state can reach, so comparing it with the interacting ground state isolates the effect of the amplitudes alone. It gives $S(L) = 0.297$ against the interacting value 0.370. Measured as an enhancement over the other sampled momenta, the classical ensemble raises L by 11% and the interacting ground state by 48%. Roughly four fifths of the L enhancement is therefore made by the quantum dynamics, not by the ice rule.

The second checks that this is not an artifact of the small cluster. Loop-update Monte Carlo on a 2048-site classical ice manifold, binned into 30 independent blocks, gives $S(L) = 0.2488(67)$, $S(X) = 0.2538(73)$, $S(W) = 0.2475(52)$ and $S(K) = 0.2467(46)$. The total spread across these zone-boundary momenta, 0.007, is comparable to the typical statistical error, 0.006: within the resolution of the calculation the classical structure factor is flat, and L in particular is not distinguished. (The zone center is excluded from this comparison; it is the uniform mode, and it carries no weight in the finite cluster.) The much stronger L enhancement seen in the interacting ground state is generated by the quantum ring dynamics at $V = 0$. This is the main distinction from an RK-point resonon, whose softening can be read directly from the classical correlations of the equal-amplitude state.

VII. MICROSCOPIC CHARACTER OF THE LOW L MODE

The state-resolved expectation values show that flippability and resonance coherence are different observables. Relative to the ground state, the low L level contains 0.34 more flippable hexagons, yet its total magnetic coherence $\sum_p \langle \cos B_p \rangle$ is smaller. Because the ring term is the full nonconstant Hamiltonian in the constrained manifold, the loss of coherence accounts directly for the excitation energy. In spin language, more plaquettes are instantaneously arranged so that they *could* flip, while the phases of those ring processes are less coherent. The mode therefore pays energy by disrupting resonance, not by violating the ice rule.

Two different single-mode energies appear in this problem and should not be confused. The Feynman ratio of Eq. (6) uses the sublattice-summed probe and gives $f_1(L)/S(L) = 0.684/0.370 = 1.85K$, which is the centroid of that state’s spectrum. A better trial state is available at no extra cost: the probe resolves into four sublattice-labelled states $S_\mu^z(L)|0\rangle$, and the lowest eigenvalue of H within that four-dimensional space is 1.450K. We take the latter as the bare single-mode energy, since

it is the best description of the excitation that uses only the electric-field wave.

The exact low L eigenstate lies at $1.024K$, so relaxation beyond the electric-field wave recovers $0.427K$, or 29% of the bare single-mode cost. Because the ring term is the entire nonconstant Hamiltonian, this recovery decomposes exactly, with no remainder, into contributions from the four hexagon families (Appendix C). Of it, 71% comes from the single family lying in the wavefronts — the family whose plaquettes carry no flux from the imposed wave and therefore contribute nothing to the oscillator strength. They can reorganize without reducing the overlap with the probe state, which is precisely the mechanism of backflow in Feynman–Cohen theory [30]: the excitation lowers its energy by rearranging the degrees of freedom the bare wave does not constrain.

VIII. FEW-PHOTON KINEMATICS

Spin-reversal parity alone does not identify the low L level. Under charge conjugation an n -photon state carries parity $(-1)^n$, so the odd parity of the level excludes even photon number but permits $n = 3$. We therefore test the three-photon assignment kinematically. Using the Gaussian branch quantized on the same torus (Appendix A), we minimize the total energy over every partition of the cluster momentum into n photon momenta drawn from the allowed set, excluding Γ , which carries no photon because the curl has no nonzero singular value there. At L the resulting thresholds are

$$\omega_L^{(1)} = 2.03K, \quad \omega_L^{(2)} = 4.03K, \quad \omega_L^{(3)} = 5.55K. \quad (7)$$

The level in question lies at $1.02K$, a factor of 5.4 below the three-photon threshold and below even the one-photon energy. The same ordering holds at every sampled momentum: the three-photon threshold never falls below $5.28K$, because the softest photon available on this cluster already costs $1.76K$. A three-photon interpretation of the L level is therefore excluded on the clusters we can diagonalize.

This argument uses the Gaussian dispersion to count the cost of the constituents, and so it is an argument about weakly renormalized photons. It does not by itself exclude a multiparticle state built from strongly renormalized constituents. What makes that alternative implausible here is that the level is a single, sharp, dominant pole carrying half of the weight at its momentum rather than the small distributed weight characteristic of a continuum, and that its polarization content is that of a single transverse branch (Appendix B).

IX. WHY THE MODE IS ROTON-LIKE

The terminology is meant to identify a mechanism and a change of character, and it is worth separating what we have established from what we have not.

We have not tracked a branch adiabatically from the Gaussian theory to the compact one, and in this model one cannot: the interpolation that would be needed does not exist. Because $E_\ell = S_\ell^z = \pm \frac{1}{2}$ makes E_ℓ^2 a constant, there is no bare electric stiffness U to turn down, hence no continuous parameter connecting a free photon to the interacting problem. Any claim of adiabatic ancestry would be rhetorical rather than computed, and we make none.

What we have established is a channel identification, which is sharper than it may sound. At each momentum the probe lives in a two-dimensional transverse space; every bright level is polarization-pure, and the two brightest are mutually orthogonal (Appendix B). The anomalous level therefore occupies one of exactly two available transverse channels. It is not an additional species beyond the photon branches, and few-photon kinematics (Sec. VIII) place it far below any multi-photon threshold.

At the same time, calling it simply a photon with a renormalized velocity misses what the state-resolved diagnostics show. The other transverse branch at L loses flippable plaquettes under the RK bias, $\Delta\langle n^{\text{flip}} \rangle = -0.06$, while this one gains them, $+0.34$; it combines increased flippability with reduced ring coherence; and its softening tracks the Feynman ratio rather than a uniform rescaling of the band, as the trajectory under V and the 64-site class test both show.

This is the sense in which the analogy to the helium roton is useful. In helium the high-momentum continuation of the sound branch is softened by correlations and backflow. Here the electric field is microscopically saturated, so a finite-amplitude wave cannot be made by increasing $|E|$ on a bond; it must instead rearrange which bonds carry the two allowed values, and that rearrangement changes the magnetic flux and the coherence of ring exchange. At L this correlated rearrangement is substantially cheaper than the Gaussian continuation of the long-wavelength photon, and the relaxation is carried by the plaquettes the wave does not constrain, which is the structure of backflow. Establishing a true local minimum of a thermodynamic dispersion would require denser momenta than our clusters provide; our claim is the finite-size, mechanism-based one stated here.

X. RELATION TO RK RESONONS

Single-mode soft modes are familiar in quantum dimer models at Rokhsar–Kivelson points. There the ground state is an equal-amplitude superposition and its static correlations are those of the corresponding classical dimer ensemble [27]. On the square lattice this produces the well-known resonon structure, confirmed in dynamical calculations [29]. The three-dimensional cubic RK model is different: its only soft mode is the remnant of the photon, consistent with the comparatively featureless classical Coulomb-phase correlations [29]. Our L anomaly

is therefore not an RK-point soft mode transplanted to pyrochlore. It occurs at the physical $V = 0$ ring point, remains at finite energy on the available clusters, and relies on an $S(L)$ enhancement generated by the interacting quantum ground state rather than by the equal-amplitude classical ensemble.

XI. FINITE-SIZE AND DEFORMATION CHECKS

Convergence of the collective state. A systematic variational improvement of the electric-field wave converges rapidly to the exact 48-site eigenvalue, $1.450 \rightarrow 1.074 \rightarrow 1.034 \rightarrow 1.026K$ against $1.024K$ from complete diagonalization. The state is reached from the probe in a few steps, as a collective mode should be, rather than requiring the full Hilbert space.

A prediction, not a fit. The 64-site cluster provides the one test in which the ground-state ratio is used in the direction it is supposed to work: forward. Its anisotropy splits the L star into two inequivalent classes, which the Gaussian theory cannot distinguish — both have the same $|\mathbf{q}|$ and the same Gaussian energy. They differ in a ground-state quantity, the equal-time structure factor: $S = 0.369$ for one class and $S = 0.227$ for the other. The f_1/S mechanism therefore predicts, before any excited state is computed, that the first class reconstructs and the second does not. It does. The class with $S = 0.369$ carries a level at $0.954K$ with 50.0% of the weight; the class with $S = 0.227$ has its dominant level at $1.998K$ with 77% of the weight, and no anomalous state at all. The effect follows the structure factor rather than the label L .

Stability with size. We can now compare three clusters. The third, $2 \times 3 \times 3$ with 72 spins, required extending the state representation past a single 64-bit word (Appendix D); it is also the first cluster here that is genuinely three-dimensional in all three directions, so it varies the shape as well as the size.

	$2 \times 2 \times 3$	$2 \times 2 \times 4$	$2 \times 3 \times 3$
sites	48	64	72
ice configurations	87 546	2 756 394	12 846 186
sector dimension	9 104	249 180	973 008
E_0/N	-0.2199	-0.2165	-0.2057
$S(L)$	0.370	0.369	0.321
ω_{low}/K	1.024	0.954	1.068
share	50.8%	50.0%	54.0%
ω_{high}/K	1.730	1.652	1.814
share	32.3%	31.3%	17.8%
$\omega_{\text{low}}/\omega_{\text{high}}$	0.592	0.578	0.589
splitting	0.513	0.535	0.518

The qualitative picture is the same on all three: a single dominant level near $1.0K$ carrying at least half of the

weight at L , and a second transverse branch roughly seventy percent higher. The two quantities that require no energy scale at all — the ratio of the two branch energies and the fractional splitting — are stable to a few percent across a factor of eleven in Hilbert-space dimension and across a change of cluster shape: 0.592, 0.578, 0.589 and 0.513, 0.535, 0.518. This is the strongest size evidence we can offer, and it is the evidence least contaminated by fitted quantities.

Not everything is stable, and we say which. The equal-time structure factor at L falls to 0.321 on the 72-site cluster from 0.370 on the other two, and the level correspondingly sits slightly higher rather than lower — the direction the f_1/S mechanism requires, though the three clusters differ in shape as well as size and we do not press this as a quantitative test. The weight retained by the upper branch drops from about a third to 18%. And the polarization splitting at X is not converged at all, falling from 53% at 48 sites to 6% at 64; no conclusion is drawn from it.

The deformation trajectory. Under a bias toward more flippable configurations ($V < 0$), $f_1(L)$ changes by less than one percent while $S(L)$ grows and the level softens from $1.02K$ to $0.73K$. The ground-state factor moves first and the spectrum follows, which is the ordering the f_1/S mechanism requires and the opposite of what a fitted correlation would produce.

XII. POLARIZATION SPLITTING AND OFF-POLE WEIGHT

The two transverse polarizations are exactly degenerate in the Gaussian theory, and their interaction-induced splitting was predicted in semiclassical treatments [22, 23]. Our calculation determines its magnitude in the microscopic ring model without an expansion. Because the two brightest levels at each momentum are polarization-pure and mutually orthogonal (Appendix B), the splitting can be read off directly as the energy difference between them. It grows monotonically as the momentum increases: it is unresolved, below the $10^{-8}K$ level-grouping tolerance, at the smallest momentum of the 64-site cluster, 4.4% at the smallest momentum of the 48-site cluster, 17% at $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$, 42% at X and 51% at L . The degeneracy is thus restored in the infrared, as the Gaussian theory requires, and destroyed at the zone boundary.

We emphasize what this does *not* add. The splitting at L and the roton-like softening at L are the same fact: the softened level is the lower of the two transverse branches, and the splitting is the size of the softening. It is reported here because it makes contact with a specific earlier prediction, not as independent evidence for the anomaly.

Beyond the two photon descendants, 16–38% of the S^z spectral weight resides in higher levels absent from the Gaussian one-photon theory. This fraction is defined without any energy window or fitted scale: the

two photon descendants are identified as the brightest level in each transverse polarization channel, and the off-pole weight is everything else (Appendix B). An alternative operational definition — simply the two brightest levels at that momentum — gives identical numbers at every momentum sampled. The off-pole fraction takes its smallest value, 16% on 48 sites and 13% on 64 sites, at the smallest momentum available on each, consistent with recovery of the infrared photon.

Several diagnostics favor a multiparticle rather than a localized magnetic interpretation of this higher weight. Plaquette-flux correlations are weak and spatially extended instead of showing the localized disturbance expected for a bound magnetic pair, and the response to the RK deformation is consistent with sums of single-mode responses. By contrast, leading-order perturbation theory in the quartic term obtained by expanding the cosine predicts polarization splittings and transferred spectral weight below 0.1% at the coupling generated by this model, and only a few percent even when that coupling is artificially tripled. The finite-momentum spectrum is therefore not quantitatively described as a weak anharmonic correction to the Gaussian photon.

XIII. PARITY AND COMPLEMENTARY PROBES

Global spin reversal leaves Eq. (1) invariant. Eigenstates can therefore be chosen even or odd under this symmetry, while S^z is odd. Consequently a spin-reversal-even ground state couples through neutron scattering only to odd states. This elementary selection rule is exact in the model; numerically every even level carries S^z weight below 10^{-28} . In gauge language it is charge conjugation, under which an n -photon state has parity $(-1)^n$.

The practical implication is a clean division between linear and bilinear probes. Even-sector weight, including the two-photon channel, is invisible to S^z but can appear in probes coupling to spin pairs, such as Raman scattering [31] and ultrasound [32]. Stray-field noise magnetometry [33], which couples to magnetization, belongs to the neutron-like odd channel. The selection rule itself is not a new symmetry statement; its value here is that the complete finite-size spectrum makes the complementary experimental channels explicit.

XIV. EXPERIMENTAL IMPLICATIONS

The calculation does not imply that every zone-boundary feature in a microscopic quantum-spin-ice material is the same as the low L level of the pure ring model. Longer ring exchanges, virtual spinons, and material-specific interactions can modify the spectrum. It does show, however, that even the minimal compact gauge Hamiltonian can redistribute a large fraction of the S^z spectral weight away from the Gaussian photon without

leaving the deconfined phase. At the sampled L point the dominant level lies at roughly half the Gaussian-photon energy and carries about half of the weight, while the other transverse branch retains only a minority. More generally, the zone-boundary response contains split photon branches and substantial off-pole intensity.

Stated as something to look for, the pattern is this. A spectrum that is a single photon pole near the zone center need not remain a single pole at the zone boundary: along [111] the weight can move onto a lower branch that sits well below the linear extrapolation of the small- q dispersion, while a weaker feature persists near where the extrapolation predicts. The two should appear at the same momentum with comparable widths and a weight ratio of roughly two to one, and the lower one should be the more intense. Because the splitting is between the two transverse polarizations, polarized neutron scattering separates them directly, which is a sharper test than the unpolarized lineshape. We caution that a finite cluster produces discrete poles where the thermodynamic system may produce a peak plus a continuum, so the robust prediction is the redistribution of weight, not the pole count.

This provides a microscopic candidate for the broad finite-momentum response seen in quantum Monte Carlo [25], but we do not identify the QMC linewidth uniquely with the finite-size levels found here. In the thermodynamic limit the gapless theory also permits multi-quantum continua to extend down toward the one-photon branch, so experimental comparison should emphasize the redistribution and lineshape of spectral weight rather than a one-to-one count of finite cluster poles.

XV. LIMITATIONS AND CLUSTERS

Our conclusions are subject to four explicit restrictions. First, the calculation is performed in the ground-state winding sector of the spin-ice ring model, not over all topological sectors and not in the full microscopic spin Hilbert space. Second, three cluster sizes are available and no more. The scale-free diagnostics — the ratio of the two branch energies and their fractional splitting — are stable across all three to a few percent, as is the level's share of the weight, but the equal-time structure factor at L and the weight retained by the upper branch are not, and neither is the splitting at X . Third, the discrete cluster momenta do not establish a local minimum of a continuous thermodynamic dispersion; this is why we use the qualified term roton-like. A zone-wide prediction would require the interacting $S(q)$ at momenta inaccessible to these clusters. Fourth, longer ring exchanges and virtual-spinon processes in microscopic XXZ models lie outside Eq. (1). Whether the anomaly survives the addition of virtual-spinon processes is a question this calculation cannot answer, and no claim here rests on it; work on the XXZ model is in progress and will be reported

separately. The overall ring scale K can be constrained thermodynamically [34].

The cluster choice itself is important. The standard 16- and 32-site pyrochlore cells are pathological for this question. On the 32-site cell every configuration in the connected zero-flux sector has exactly eight flippable hexagons, making the equal-amplitude state an exact eigenstate and producing an artificially integer-valued spectrum; on the 16-site cell only six transport sectors have any dynamics. We therefore use the $2 \times 2 \times 3$ cluster (48 sites, 87 546 ice configurations; ground-state winding sector of dimension 9104), which we diagonalize completely so that the operator-resolved weights exactly saturate the equal-time sum rule. For the size checks we use $2 \times 2 \times 4$ (64 sites, 2 756 394 ice configurations; ground-state winding sector of dimension 249 180) and $2 \times 3 \times 3$ (72 sites, 12 846 186 configurations; sector dimension 973 008), both by Lanczos and both only after reproducing all 48-site benchmarks. Momentum labels are certified by exact translation projectors.

It is natural to ask why we do not simply vary the *shape* of the torus at fixed spin number, which would be the cleanest control of all. The reason is that at these sizes there is no second admissible shape. A cluster is admissible only if it carries the full complement of contractible hexagons, one per site, as the infinite lattice does. The $2 \times 2 \times 3$ and $2 \times 3 \times 3$ tori do: they have 48 hexagons on 48 sites and 72 on 72. Every other tiling of 48 or 64 sites that contains an L -type momentum is one unit cell thick in some direction — $1 \times 3 \times 4$, $1 \times 2 \times 6$, $3 \times 4 \times 1$ and their 64-site analogues — and in those the majority of six-cycles close only by winding around the torus. The $1 \times 3 \times 4$ cluster, for instance, retains just 24 contractible hexagons on 48 sites. Keeping only the contractible ones leaves a Hamiltonian with half the ring terms per site, which is a different model rather than the same model in a different box. Shape variation at fixed size is therefore not an available test here, and the finite-size evidence rests on the sequence of sizes instead.

XVI. CONCLUSION

The standard pyrochlore ring Hamiltonian is known to realize a strongly interacting compact $U(1)$ gauge theory; our result concerns how that strong coupling appears in its operator-resolved finite-momentum spectrum. At long wavelength the neutron response is dominated by the expected emergent photon. At the sampled L point, however, the spectral weight reconstructs around a collective level at about half the Gaussian-photon energy carrying approximately half of the intensity. Its connection to a transverse photon branch, its softening with f_1/S , its positive flippability response, and the recovery of energy through wavefront plaquettes motivate a Feynman-type, roton-like interpretation. The same level is the lower of the two transverse branches, so the order-one polarization splitting at the zone boundary is not a second result

but the same reconstruction expressed as an energy difference; it connects what we find to earlier semiclassical predictions [22, 23]. Separately from the photon descendants, 16–38% of the S^z spectral weight lies in levels the Gaussian one-photon theory does not contain. The principal experimental message is therefore not that the emergent photon disappears, but that finite-momentum spectroscopy of a compact spin-one-half gauge field need not resemble a single renormalized photon pole. single renormalized photon pole.

Appendix A: The Gaussian benchmark

Every statement of the form “about half the Gaussian-photon energy” refers to the following construction, which is quantized on the same finite torus as the exact diagonalization so that no continuum extrapolation enters the comparison.

Write $B_p = \sum_{\ell} C_{p\ell} A_{\ell}$, where C is the alternating lattice curl: $C_{p\ell} = \pm 1$ with alternating sign around the six links of hexagon p , and zero otherwise. Expanding the compact magnetic energy of Eq. (1) to quadratic order and adding the electric term gives

$$H_G = \frac{U}{2} \sum_{\ell} E_{\ell}^2 + K \sum_p B_p^2, \quad [A_{\ell}, E_{\ell'}] = i\delta_{\ell\ell'}. \quad (\text{A1})$$

Restricting to momentum $\mathbf{q}$ and using the four-dimensional sublattice Bloch basis, the normal-mode frequencies are set by the singular values $\sigma_{\lambda}(\mathbf{q})$ of the curl in that block,

$$\omega_{\lambda}(\mathbf{q}) = \sqrt{2UK} \sigma_{\lambda}(\mathbf{q}). \quad (\text{A2})$$

The construction is checked, not assumed. (i) Gauge invariance: $C d_0 = 0$ to machine precision, where d_0 is the tetrahedron incidence matrix signed by the up/down bipartition; the largest element of $C d_0$ is 0 exactly. If this fails the sign convention is wrong and nothing downstream is meaningful. (ii) At every $\mathbf{q} \neq \Gamma$ the block has exactly two nonzero singular values, the two transverse photon polarizations, and at Γ it has none. (iii) The Bloch bases are orthonormal, and the probe at $+\mathbf{q}$ pairs with modes at $-\mathbf{q}$, verified by projection completeness.

A single number is fitted: the overall energy scale $\sqrt{2UK}$. In the pure ring model U is generated dynamically rather than being a bare coupling, so its value is not available from Eq. (1); this is the one place where we import a scale from the data. It is determined by one linear least-squares fit of the dominant exact line against the dominant Gaussian branch across all momenta, giving $\sqrt{2UK} = 0.5868 K$. Everything else — the *shape* of the dispersion across momenta, the degeneracy or splitting of the two polarizations at each $\mathbf{q}$, and the distribution of one-photon weight over momenta and sublattice channels — is parameter-free geometry. The resulting values are those plotted as the dashed Gaussian curve in Fig. 1 and used as the denominator in Fig. 2(a); at L , $\omega_G = 2.03K$.

We stress the limitation this creates. Because one scale is fitted to the exact spectrum, the Gaussian curve cannot be used to test whether the overall photon velocity is renormalized: by construction it is not. It can only be used for statements that are invariant under a uniform rescaling of the whole branch — which momenta are anomalous *relative to the others*, and whether the two polarizations split. Those are the only statements we draw from it.

Appendix B: Polarization resolution of the bright levels

At each momentum the electric field lives in the four-dimensional sublattice space, in which the lattice curl has exactly two nonzero singular values. The neutron probe is therefore confined to a two-dimensional transverse subspace; the weight found in the remaining two directions is the longitudinal residue, which must vanish because $\nabla \cdot \mathbf{E} = 0$ in the ice manifold. It does, at the level of 10^{-31} relative to $S(\mathbf{q})$ at every momentum. This is an exact gate on the decomposition, not a fit.

A caution is necessary before any polarization statement is made. The two Gaussian transverse branches are *exactly* degenerate at every momentum, so the singular-value decomposition determines the two polarization directions only up to an arbitrary rotation within that subspace. A statement such as “this level is 68% polarization 1” is consequently meaningless in isolation. The invariant content is the geometry of the levels’ polarization vectors relative to one another. For a level n of multiplicity m we form the 2×2 polarization density matrix $M_n = A_n^\dagger A_n$, where A_n is the $m \times 2$ matrix of transverse amplitudes $\langle n, \alpha | S^z(\mathbf{q}) | 0 \rangle$. Both the purity $\text{tr } M_n^2 / (\text{tr } M_n)^2$ and the normalized overlap $\text{tr}(M_A M_B) / (\text{tr } M_A \text{tr } M_B)$ between two levels are invariant under rotations of the degenerate subspace.

The result is unambiguous. Every bright level, including multiplets of multiplicity up to four, has polarization purity 1.000000: the whole multiplet couples through a single polarization direction. And the two brightest levels at each momentum are mutually orthogonal in polarization, so a basis exists in which each is pure. They are therefore the two transverse photon descendants, identified without reference to any energy scale.

$\mathbf{q}$	ω_1/K	ω_2/K	splitting	overlap
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	1.895	1.980	0.044	2×10^{-28}
$(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$	1.744	2.069	0.171	1×10^{-29}
$(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$	2.533	2.687	0.059	4×10^{-2}
L	1.024	1.730	0.513	3×10^{-3}
X	2.235	3.423	0.420	1×10^{-31}

The splitting column is $|\omega_1 - \omega_2| / \frac{1}{2}(\omega_1 + \omega_2)$. Orthogonality is exact to machine precision at $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ and X , where the little group forbids mixing, and holds

to 0.3% at L and 4% at $(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$, where it does not. The residual mixing is small enough that the branch assignment is never in doubt.

Two consequences matter for the interpretation. First, the identification of the anomalous L level as a photon descendant, and not a new species, rests on this decomposition and is independent of the fitted Gaussian scale. Second — and this must be stated plainly — the “interaction-induced polarization splitting” and the “roton-like softening” at L are not two independent findings. They are the same fact viewed twice: the lower of the two transverse branches at L is the anomalous level, and the splitting between the branches is the amount by which it has been pushed down. We report the splitting because it connects to a specific earlier prediction [22, 23], not as separate evidence.

Appendix C: Exact decomposition of the backflow energy

Because the ring term is the entire nonconstant Hamiltonian inside the ice manifold, the excitation energy of any state $|n\rangle$ is exactly a sum of plaquette contributions,

$$\omega_n = -K \sum_p \left[\langle n | W_p + W_p^\dagger | n \rangle - \langle 0 | W_p + W_p^\dagger | 0 \rangle \right], \quad (\text{C1})$$

with no remainder. Grouping the plaquettes by hexagon family — the four $\langle 111 \rangle$ orientations, labeled unambiguously by the sublattice each hexagon omits, which is immune to the periodic wrapping that corrupts plane fitting on a small torus — turns Eq. (C1) into a four-term decomposition of any energy. Applying it to the bare single-mode state — the lowest eigenvector of H within the four-dimensional space spanned by the sublattice-resolved probe states $S_\mu^z(L)|0\rangle$, whose energy is $1.450K$ — and to the exact lowest L eigenstate at $1.024K$ gives, in units of K ,

family	1	2	3	4	total
single mode	0.422	-0.540	1.004	0.565	1.450
exact	0.341	-0.476	0.702	0.456	1.024
relief	0.080	-0.064	0.301	0.110	0.427

where family 3 is the one with $g_p(L) = 0$ identically. The totals reproduce the variational and exact energies, so the decomposition is complete rather than approximate. Of the $0.427K$ of energy recovered in going from the single-mode state to the exact eigenstate, $0.301K$, or 71%, comes from the dark family alone — although it constitutes one quarter of the plaquettes and contributes nothing to the oscillator strength. This is the precise sense in which the relaxation is backflow-like: the state reorganizes exactly those degrees of freedom that the bare electric-field wave does not constrain, and which therefore cost it no overlap.

Appendix D: Numerical methods

Ice configurations are enumerated exactly by depth-first assignment, one site at a time, backtracking as soon as any tetrahedron would acquire a third spin of either sign, and are stored as bit masks. The ring Hamiltonian is built sparsely: for each hexagon a bit mask and the two flippable bit patterns identify the acted-on states in one vectorized pass, and the images are located by binary search in the sorted state list. The winding sector is identified from the polarization $\sum_i (n_i - \frac{1}{2}) \mathbf{r}_i$, which is conserved by Eq. (1); we retain the sector carrying the ground state.

On the 48-site cluster the ground-state winding sector has dimension 9104 and is diagonalized completely with a dense symmetric eigensolver, so all 9104 eigenpairs are available and the operator-resolved weights saturate the equal-time sum rule $\sum_n |\langle n | S_\mu^z(\mathbf{q}) | 0 \rangle|^2 = S_\mu(\mathbf{q})$ exactly rather than to a tolerance. On the 64-site cluster the sector has dimension 249 180 and is treated by Lanczos with full reorthogonalization, the dynamical response being obtained by a continued-fraction expansion seeded with $S_\mu^z(\mathbf{q}) | 0 \rangle$; all 48-site benchmarks are reproduced before any 64-site number is quoted.

Going beyond 64 spins requires changing the state representation. Storing a configuration in a single 64-bit word caps the calculation at 64 sites, which is where the published constrained-space spectroscopy has stood. We therefore store each configuration as a *pair* of 64-bit words and redefine masking, flipping, sorting and lookup on that pair. Lookup is the only part that is not immediate: the table is sorted by the pair lexicographically, and because the high word takes fewer than 2^8 distinct values at these sizes, queries are grouped by high word and each group resolved with a single vectorized binary search on the low word within its block. The result is exact, not hashed. The implementation is validated by running it on the 48-site cluster, where it reproduces the 64-bit

pipeline exactly: $E_0 = -10.55601176K$, $S(L) = 0.37036$ and branch energies $1.0235K$ and $1.7303K$.

On the 72-site cluster the ice manifold contains 12 846 186 configurations in 310 winding sectors. The ground state lies in the zero-polarization sector, of dimension 973 008; we verify this rather than assume it, by diagonalizing the largest sectors independently and comparing, which also exposes the expected exact degeneracy between symmetry-related sectors ($E_0 = -14.80971536K$ in two of them). The dynamical response is again obtained by continued fraction seeded with the four sublattice components of the probe. Because those four seeds produce separate lines at the same energy, they are summed by energy before any branch energy or weight is quoted; only the channel sum is basis-independent.

Momentum labels are certified by explicit translation projectors rather than inferred from position in a sorted list. Degenerate levels are grouped with a tolerance of $10^{-8}K$ and their weights summed, since only the total weight of a degenerate multiplet is basis-independent.

Three gates are applied throughout and reported with the data: the plaquette sum rule $\sum_p \langle W_p + W_p^\dagger \rangle = -E_0/K$, satisfied to 10^{-6} ; the exactness of Eq. (5) against the directly computed spectral first moment, ratio 1.0000; and the gauge identity $C d_0 = 0$ of Appendix A. The scripts that produce every number and figure in this paper are included with the submission.

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We thank the Digital Research Alliance of Canada for computational resources. The complete set of scripts that generates every number, gate and figure reported here is included with the submission.

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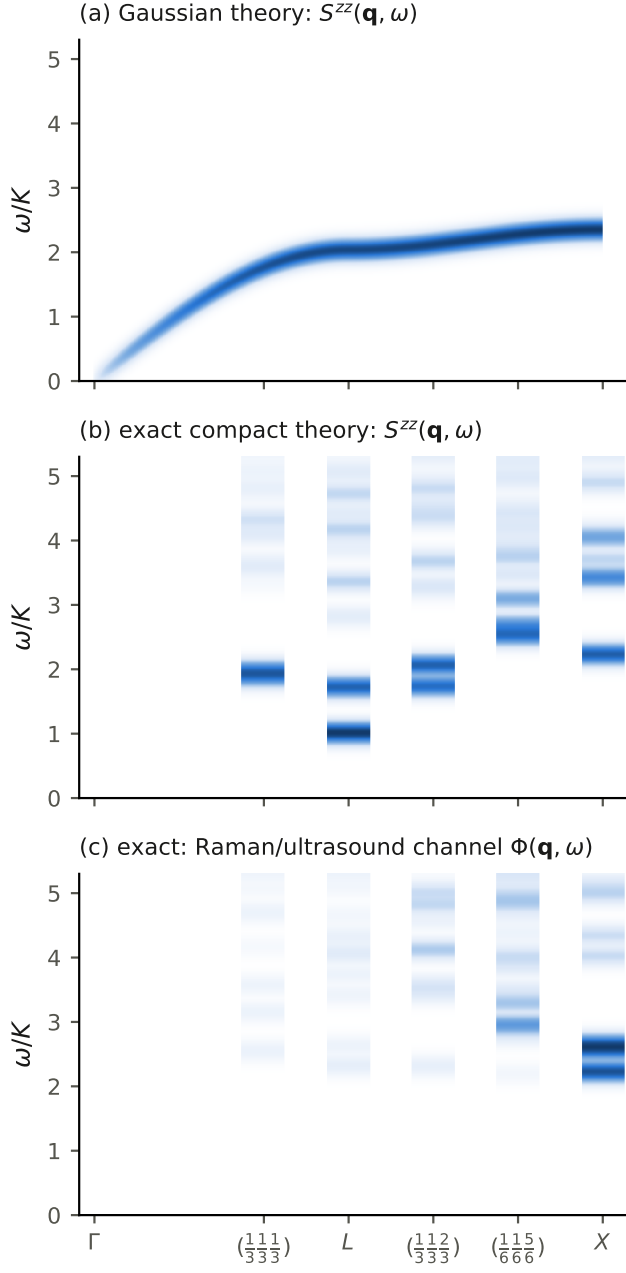


FIG. 3. Simulated spectra along Γ - L - X for the 48-site cluster, broadened by $\eta = 0.08K$ and normalized separately in each panel. (a) Gaussian one-photon neutron response. (b) Exact operator-resolved S^z response within the ground-state winding sector: split photon descendants, a dominant low L mode at about half the Gaussian-photon energy, and higher off-pole weight. (c) Spin-reversal-even response accessible to bilinear probes. By the parity selection rule no eigenstate is shared between panels (b) and (c).